

QTM 220 Final Exam (12/7)

Problem 1

Suppose you have a sample $Y_1, Y_2, \dots, Y_n$ drawn with replacement from a population with mean μ and standard deviation σ .

Consider the following estimator for μ : $\tilde{Y} = \frac{1}{2n} \sum_{i=1}^n Y_i$

Part A

What is $E[\tilde{Y}]$?

$$E[\tilde{Y}] = E\left[\frac{1}{2n} \sum_{i=1}^n Y_i\right]$$

$$= \frac{1}{2n} E\left[\sum_{i=1}^n Y_i\right]$$

$$= \frac{1}{2n} \sum_{i=1}^n E[Y_i]$$

by linearity of expectations

$$= \frac{1}{2n} \sum_{i=1}^n E[Y]$$

by IID

$$= \frac{1}{2n} \sum_{i=1}^n \mu$$

by definition

$$= \frac{1}{2n} n \mu$$

$$= \frac{n}{2n} \mu$$

$$= \frac{1}{2} \mu$$

Part B

How does $E[\tilde{Y}]$ compare to $E[\bar{Y}]$? (where $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$)

similar steps show that $E[\bar{Y}] = \mu$. $\bar{Y}$ is unbiased, $\tilde{Y}$ is biased; bias does not change in n

Part C

Showing your work, calculate $Var[\tilde{Y}]$.

$$Var[\tilde{Y}] = V[\frac{1}{2n} \sum_{i=1}^n Y_i]$$

$$= (\frac{1}{2n})^2 V[\sum_{i=1}^n Y_i]$$

$$= (\frac{1}{2n})^2 \sum_{i=1}^n V[Y]$$

because IID, covariances are 0

$$= (\frac{1}{2n})^2 \sum_{i=1}^n \sigma^2$$

by definition

$$= (\frac{1}{2n})^2 n \sigma^2$$

$$= \frac{n}{(2n)^2} \sigma^2$$

$$= \frac{1}{4n} \sigma^2$$

Part D

How does $Var[\tilde{Y}]$ compare to $Var[\bar{Y}]$?

Similar steps show that $V[\tilde{Y}] = \frac{\sigma^2}{n}$

$\frac{1}{n} > \frac{1}{4n}$ so our $\tilde{Y}$ estimator has a lower variance than the sample mean. But the estimator is biased, so we probably wouldn't want to use it over the mean.

Problem 2

Suppose we have a sample of $Y_1 \dots Y_n$ and binary $X_1 \dots X_n$ stored in an **R** dataframe 'df' with n rows. We run this code.

```
B=10000
theta<- rep(NA,B)
for( b in 1:B ){
  bootind = sample(1:n,size=n,replace=T)
  dfboot = df[bootind,]
  theta[b]<- (mean(dfboot$y[dfboot$x==1])-mean(dfboot$y[dfboot$x==0]))
}
```

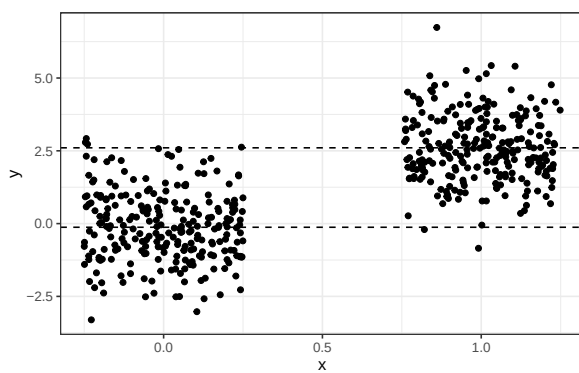
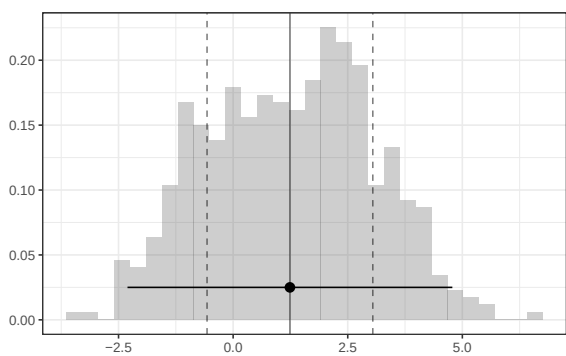


Figure 1: Histogram of the vector Y 's elements Figure 2: Jittered Scatterplot of X and Y

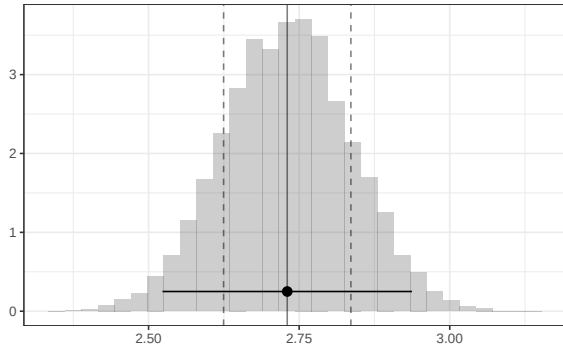


Figure 3: Histogram of the theta's elements

Part a

What does this code calculate? State it in words and mark it with an 'a', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mean(Y)
```

The sample mean. It's indicated by the solid vertical line on the sample histogram.

Part b

What does this code calculate? State it in words and mark it with a 'b', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(Y)
```

The sample standard deviation of Y. It's the distance between the solid vertical and either dashed vertical line on the sample histogram.

Part c

What does this code calculate? State it in words and mark it with a 'c', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mean(Y[X==0])
```

The conditional sample mean of Y among X=0. It is the lower dashed line on the scatterplot.

Part d

What does this code calculate? State it in words and mark it with a ‘d’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(theta)
```

The bootstrap sampling distribution’s standard deviation for the difference in the mean of y between x=1 and x=0. Estimated standard error. It’s the distance between the solid vertical and either dashed vertical line on the theta sampling distribution histogram.

Part e

What does this code calculate? State it in words and mark it with a ‘e’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
(mean(y[x==1])-mean(y[x==0])) + 1.96*(sd(y[x==1])/sqrt(n1) + sd(y[x==0])/sqrt(n0))  
(mean(y[x==1])-mean(y[x==0])) - 1.96*(sd(y[x==1])/sqrt(n1) + sd(y[x==0])/sqrt(n0))
```

A 95% confidence interval — based on normal approximation — for the difference in means for the population from which this sample was drawn. This is indicated by the horizontal segment (with a dot in the middle) on the sampling distribution histogram (theta).

Part f

How could you calculate roughly the same thing as in part e using ‘sd(theta)’

We can use the standard deviation of the bootstrap sampling distribution as our estimate of standard error.

```
(mean(y[x==1])-mean(y[x==0])) + 1.96 * sd(theta)  
(mean(y[x==1])-mean(y[x==0])) - 1.96 * sd(theta)
```

conceptual addition: why no n? What is standard error? talk through bootstrap in words?

Problem 3

change to quadratic!

Problem 4

We run an experiment in which a treatment, D , is randomly assigned to participants conditional on sex. The table below shows our data, including our outcome of interest, Y .

Sex	Treatment (D)	Outcome (Y)	Y(1)	Y(0)
Male	1	20		
Male	0	5		
Male	0	10		
Male	0	5		
Female	1	10		
Female	1	5		
Female	0	5		
Female	0	0		
Female	0	5		
Female	1	10		
Female	0	5		
Female	1	5		

Part A

Fill in the table with potential outcomes based on the observed data. Put a question mark (?) in the cells for which you do not have data.

Part B

What is the probability of receiving treatment for males? What is the probability of receiving treatment for females?

$$P(D = 1|Male) = \frac{1}{4}$$

$$P(D = 1|Female) = \frac{1}{2}$$

Part C

What is the treatment effect among males? What is the treatment effect among females?

$$\text{Males: } 20 - \frac{5+10+5}{3} = 20 - \frac{20}{3} = 13.333$$

$$\text{Females: } \frac{10+5+10+5}{4} - \frac{5+0+5+5}{4} = \frac{30}{4} - \frac{15}{4} = 7.5 - 3.75 = 3.75$$

Part D

If you ignored the conditional randomization and calculated the average treatment effect without accounting for sex, what would your estimate be?

$$\frac{20+10+5+10+5}{5} - \frac{5+10+5+5+0+5+5}{7} = \frac{50}{5} - \frac{35}{7} = 10 - 5 = 5$$

Part E

To analyze our data, we are interested in fitting the following model:

$$Y_i = \beta_0 + \beta_1 D_i + \beta_2 Sex_i + \beta_3 D_i : Sex_i$$

where $Sex_i = 1$ for males and $= 0$ for females. Fill in the table below using β s

	β s
Avg Y [Treated, Male]	$\beta_0 + \beta_1 + \beta_2 + \beta_3$
Avg Y [Control, Male]	$\beta_0 + \beta_2$
Avg Y [Treated, Female]	$\beta_0 + \beta_1$
Avg Y [Control, Female]	β_0
$\hat{ATE}_{Male}$	$\beta_1 + \beta_3$
$\hat{ATE}_{Female}$	β_1

conceptual addition: when we run this regression in R, why might we be concerned about using the standard errors `lm()` gives us? [homoskedasticity assumption]

Problem 5

leave one out cross-validation but with a quadratic

Problem 6

piece-wise constant

Problem 7

omitted variable bias with interaction

Part A

Suppose we want β_1 from $y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 x_i : z_i + e_i$ but we run the regression $lm(y \sim x)$ without z or the interaction term. We have de-meaned our variables x_i and z_i such that $\bar{x} = \sum_{i=1}^n x_i = 0$ and $\bar{z} = \sum_{i=1}^n z_i = 0$. Derive what your coefficient on x from $lm(y \sim x)$, which is $\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$, will be in terms of β s and justify your steps. Hint: you can relabel $x_i : z_i$ as a different variable.

The coefficient on x from $lm(y \sim x)$ is

$$\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$$

Plug in y_i defined by the multivariate regression: $y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 x_i : z_i + e_i$

Let's redefine $x_i : z_i = w_i$

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 w_i + e_i$$

$$\frac{\sum_{i=1}^n x_i (\beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 w_i + e_i)}{\sum_{i=1}^n x_i^2}$$

Distribute

$$\frac{\sum_{i=1}^n x_i \beta_0}{\sum_{i=1}^n x_i^2} + \frac{\sum_{i=1}^n x_i \beta_1 x_i}{\sum_{i=1}^n x_i^2} + \frac{\sum_{i=1}^n x_i \beta_2 z_i}{\sum_{i=1}^n x_i^2} + \frac{\sum_{i=1}^n x_i \beta_3 w_i}{\sum_{i=1}^n x_i^2} + \frac{\sum_{i=1}^n x_i e_i}{\sum_{i=1}^n x_i^2}$$

Let's work through each term.

$$\frac{\sum_{i=1}^n x_i \beta_0}{\sum_{i=1}^n x_i^2} = \frac{\beta_0 \sum_{i=1}^n x_i}{\sum_{i=1}^n x_i^2} = \frac{\beta_0 * 0}{\sum_{i=1}^n x_i^2} = 0$$

as x is demeaned, $\sum x_i = 0$

$$\frac{\sum_{i=1}^n x_i \beta_1 x_i}{\sum_{i=1}^n x_i^2} = \frac{\beta_1 \sum_{i=1}^n x_i x_i}{\sum_{i=1}^n x_i^2} = \beta_1$$

$$\frac{\sum_{i=1}^n x_i \beta_2 z_i}{\sum_{i=1}^n x_i^2} = \frac{\beta_2 \sum_{i=1}^n x_i z_i}{\sum_{i=1}^n x_i^2}$$

$$\frac{\sum_{i=1}^n x_i \beta_3 w_i}{\sum_{i=1}^n x_i^2} = \frac{\beta_3 \sum_{i=1}^n x_i w_i}{\sum_{i=1}^n x_i^2}$$

$$\frac{\sum_{i=1}^n x_i e_i}{\sum_{i=1}^n x_i^2} = \frac{0}{\sum_{i=1}^n x_i^2} = 0$$

we know that $\sum_{i=1}^n x_i e_i = 0$ from the first order conditions on our least squares estimator (we chose the intercept and coefficient(s) that minimize the sum of squared residuals)

Thus we have

$$\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2} = \beta_1 + \frac{\beta_2 \sum_{i=1}^n x_i z_i}{\sum_{i=1}^n x_i^2} + \frac{\beta_3 \sum_{i=1}^n x_i w_i}{\sum_{i=1}^n x_i^2}$$

Plugging back in for w_i ,

$$\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2} = \beta_1 + \frac{\beta_2 \sum_{i=1}^n x_i z_i}{\sum_{i=1}^n x_i^2} + \frac{\beta_3 \sum_{i=1}^n x_i z_i x_i}{\sum_{i=1}^n x_i^2}$$

conceptual question: what happens if x is randomized?